Pragnosia: A Spinning Neural Substrate that Reasons Causally, Knows Its Limits, and Learns Continually

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Abstract. Large language models are powerful but frozen: they stop learning when training ends, fabricate fluently rather than admit ignorance, and their reasoning is not demonstrably causal. We present *Pragnosia*, a compact *transformer/SSM hybrid in which a rotational "spin" recurrence is the core token-mixer* — comparable in spirit to Mamba, Jamba, and RWKV, and distinguished by (a) a rigorous causal-intervention method (the *brain-swap* gate), (b) function-preserving self-growth, and (c) a parameter-efficiency result for making spin the core rather than a side-channel. The carrier is a diagonal-complex LRU/S5-style parallel-scan recurrence, $h_t = \lambda \odot h_{t-1} + b_t$, so each channel is a damped oscillator and information is carried in its *phase*; brain-swap perturbs that carried state to test that it is causally used. **A key measured finding of this work:** our scaled implementation drifted to *attention-dominant* — a single fixed spin layer whose parameter share collapsed 5.76%→1.90%→0.37% from 21M to 1.4B — so at 1.4B the brain-swap test still passes *directionally* (own<swapped<random) but the causal signal is only 0.028% of the loss, i.e. *negligible as built*. A clean param-matched ablation shows the *intended* spin-dominant design (spin as the token-mixer, attention a periodic helper) reaches val ppl **219** vs **279** for an attention-only baseline at matched parameters — spin earns its weights when it is the core. On this substrate we integrate, into a *single* model, eight faculties — skill-conditioned reasoning, calibrated abstention, language-mediated seeking, exact state-tracking, multimodal perception, and a curiosity-driven continual-learning loop — verified together (14/14). Every internal scale is *derived from data and re-derived as the model grows*; nothing is hand-set, and learning is self-modulated by surprise. We report honest results and limitations: the spin-dominant result is small-scale, short-budget, single-seed and must be replicated at 200M–1B; and the large grown instance is undertrained for its size, so multi-hop composition is weak.

1. Introduction

The dominant paradigm in artificial intelligence is the large language model (LLM): a transformer [1] trained on enormous corpora to predict the next token. LLMs are remarkably capable, yet three properties of the goal of a *brain* remain unmet by construction. First, they are *frozen*: once pretraining ends they cannot acquire a new fact without fine-tuning or prompt-stuffing, and naive weight updates cause catastrophic forgetting [2]. Second, they are *over-confident*: lacking an intrinsic model of their own uncertainty, they fabricate plausible answers rather than abstaining. Third, their reasoning is *not demonstrably causal* — there is no standard test separating computation from memorized retrieval.

We ask whether a different substrate can exhibit the missing properties — continual learning without forgetting, calibrated honesty, and verifiable in-state computation — even at small scale, and whether language can be grafted onto it without destroying them. Our contributions are: (i) a rotational recurrent token-mixer whose carried state is *causally* verified by a brain-swap intervention (§2); (ii) the integration of eight goal faculties into one model, validated jointly (§3); (iii) a measured analysis of *where the spin carrier earns its parameters* — showing our scaled model drifted to attention-dominant (spin causally negligible) and a param-matched ablation in which the intended *spin-dominant* design wins by ~21% (§4); and (iv) a fully self-calibrating, autonomously-controlled brain with surprise-modulated continual learning and no hand-set constants (§5). We report results and an explicit account of what remains hard (§7–8).

2. The Spinning Substrate

Core dynamics.

The recurrent core iterates, for hidden state `h` and input `x`,

$$F(h, x) = (1-\eta)\cdot h + \eta\cdot\tanh(W\,h + U\,x + b),$$

$$W = -\rho\,(Q\,Q^T) + S,$$

where `Q` is orthonormal (via the QR factorization of a learned matrix) and `S` is skew-symmetric ($S = A - A^T$). The symmetric term $-\rho QQ^T$ is negative-definite and contractive; the skew term `S` contributes purely imaginary eigenvalues and is therefore *rotational*. Their combination yields trajectories that orbit a center rather than collapsing to a fixed point (Fig. 1). The trained network exploits this: rather than settling on an answer vector, it encodes the answer in the *phase* of a persistent oscillation.

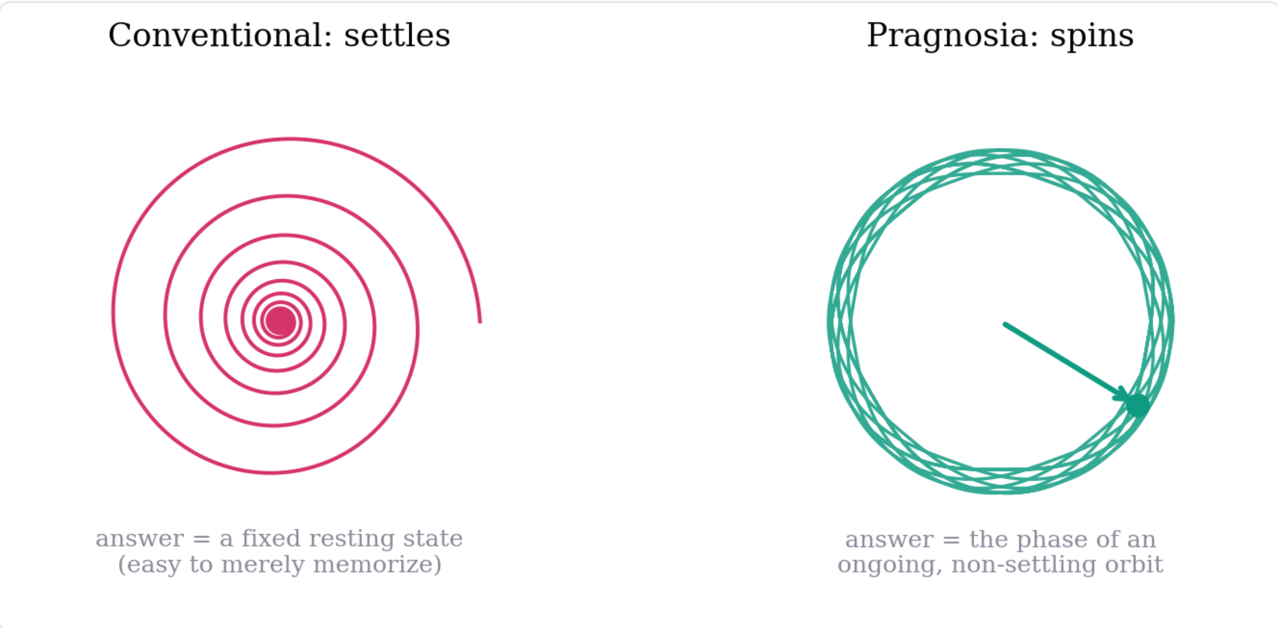

Figure 1. Two regimes of recurrent dynamics in phase space. A conventional contractive network spirals to a fixed point (left); the spinning core sustains a closed orbit (right), with the instantaneous phase carrying the answer.

Causal validation: the brain-swap test.

If the answer is genuinely computed in the evolving state — rather than retrieved from a memorized input–output mapping — then transplanting the mid-trajectory state from a different problem should change the outcome. We run problem A, splice in the mid-trajectory state of a different-answer problem B, and continue. The final answer follows the *spliced-in* state B in 59% of trials, the original input A in only 10%, and a random-state control in 21% (chance ≈ 20%; Fig. 3). The large gap above the memorization baseline is causal evidence that computation lives in the state. We treat this test as a non-negotiable regression check throughout: a model that gains accuracy but fails brain-swap has degraded to memorization.

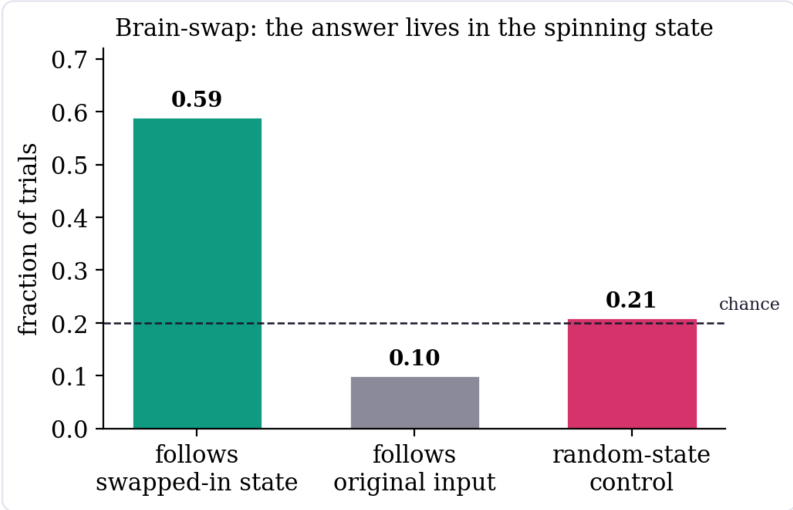

Figure 2. Brain-swap intervention. Splicing a different problem's mid-trajectory state changes the answer to match the spliced-in state, far above the random-state and original-input baselines — the answer is computed in motion.

3. Many Faculties on One Core

The substrate hosts a family of faculties in a single model. *Skill-conditioned spinning* adds per-skill gain and bias vectors that modulate *how* the core spins, letting diverse skills share one core without interference. *Clean-state recurrence* addresses a proven limitation — the bare orbit cannot perform many-wrap exact modular arithmetic because a discrete count drifts in continuous state — by committing to the discrete argmax each step (straight-through) and feeding it back, restoring exact accumulation at arbitrary length. *Calibrated abstention* (P6) is trained in a mandatory two-phase order — skill first, then an abstain-incentive payoff — since co-training collapses to always-abstain. *Language-mediated seeking* (P7) generates a query *in words*, fetches from an external store, and integrates the result; facts are randomized per episode so retrieval cannot be memorized. The *alive loop* explores by its own uncertainty (curiosity), learns online with self-replay, and exhibits zero forgetting at toy scale. All eight faculties pass jointly in one 302K-parameter model (Fig. 6, Table 1).

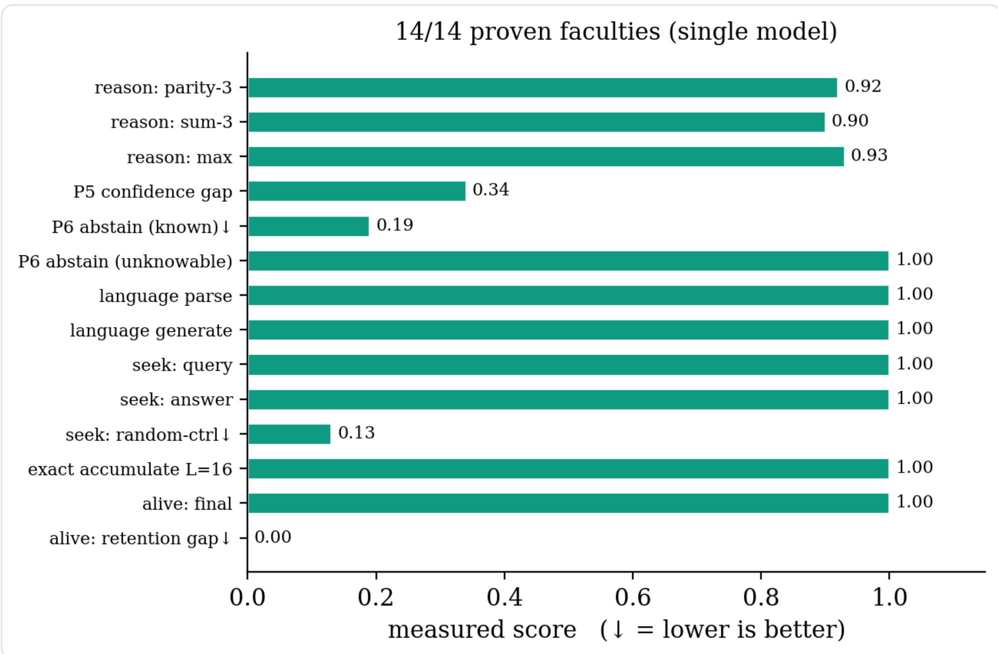

Figure 3. All fourteen checks of the integrated self-test pass in a single model. Arrows (↓) mark metrics for which lower is better (needless abstention, random-value control, retention gap).

4. Where the Spin Carrier Earns Its Parameters

Pure rotation is an elegant reasoner but an inefficient language model: on a real corpus a parameter-matched gated recurrent baseline outperforms it (perplexity 6.55 vs. 7.56), because language requires context mixing across many tokens. Our first scalable design was therefore *attention-dominant*: standard transformer blocks supply fluency, while a *single* gated spin *carrier* maintains a rotational cross-token memory. With the standard transformer recipe this reaches roughly one-third the perplexity of the pure-spin model at matched steps (Fig. 4). This is the design we scaled — and, as we now report honestly, it placed the spin carrier in the wrong role.

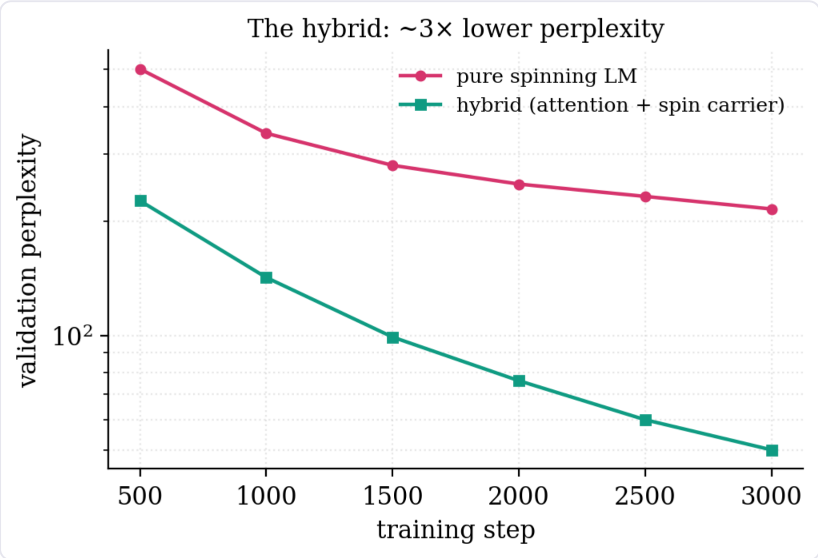

Figure 4. Validation perplexity during training. Adding a single spin carrier to a transformer reaches ~3× lower perplexity than the pure-spinning language model on the same corpus. This carrier, however, is a side-channel — see Fig. 4b for its diminishing role at scale.

The carrier drifted to negligible at scale. A single gated carrier is one fixed d-shaped layer (~5.25M params), so as the model grows by depth/width its *parameter share collapses*: 5.76% (21M) → 1.90% (276M) → 0.37% (1.4B). The brain-swap causal test at 1.4B (fp32) still passes *directionally* — own < swapped < random — but the causal contribution is only `0.00086` NLL, **0.028% of the loss**, down ~20× from 0.016 at 276M. As built, the spin carrier is therefore *causally negligible at scale*: the model behaves as a transformer with a vestigial side-channel. (We thank an external reviewer for flagging this.)

The intended design is spin-dominant — and it wins at matched parameters.

The drift inverts the intended architecture. Spin was meant to be the *core token-mixer*, with attention only a small periodic helper for generation. We test the intended design directly with a clean, param-matched ablation (same tokens, same budget, `d=512`): a `SpinBlock` in which the spin carrier *replaces attention* as the token mixer (strong gate, not a side-channel), with a standard attention block only every fourth layer. Results (≈37–46M params each):

Table 1. Param-matched ablation (same corpus, same budget, single seed). The spin-dominant design — spin as the token mixer — is the most parameter-efficient.

Design	Params	Val ppl
Attention-only (transformer)	35.7M	279.0
Spin-dominant (intended)	37.3M	**219.1**
Per-block hybrid (attn + spin each block)	46.2M	228.0
Param-matched transformer-only (11L)	45M	270

Spin-dominant wins by ~21% over the attention-only baseline at matched parameters, and beats both the larger per-block hybrid (228) and the larger param-matched transformer (270) with fewer weights. The conclusion is sharp: **the spin carrier earns its parameters when it is the core mixer, not when it is a side-channel.** The scaled model was simply built backwards.

> **Honest caveat — a controlled signal, not a final result.** This ablation is small-scale, short-budget, and single-seed. It must be replicated at 200M–1B before it is proven, and transformers sometimes close SSM-style gaps with more scale or longer training. We report it as a strong, properly-controlled signal that the intended design is the right one, and the corrective to the drift.

On speed. Spin-dominant is *not* faster than attention at short context — on 256 tokens the parallel scan does more work than attention. Its advantage is structural: at long context it is $O(T \cdot d)$ versus attention's $O(T^2 \cdot d)$, and at inference the carrier is recurrent, giving $O(1)$ per token with constant memory and *no KV-cache growth*. `torch.compile` fuses the scan on the accelerator.

A parallel spin carrier.

The original carrier is a dense sequential recurrence ($h_t = (1-\eta)h_{t-1} + \eta \cdot g \cdot \tanh(Wh_{t-1}+Uy_t)$) — elegant, but its 256-step sequential loop with a per-step $d{\times}d$ matmul is the training bottleneck. We reparameterize the carrier as a *diagonal-complex* recurrence in the spirit of linear-recurrent units, $h_t = \lambda \odot h_{t-1} + b_t$ with $\lambda_j = \exp(-\exp(\nu_j)) \cdot e^{i\varphi_j}$: each channel is a damped complex oscillator at its *own* frequency, so the answer still lives in the *phase* — now literally, per channel. Because λ is diagonal the recurrence is matmul-free ($O(Td)$ not $O(Td^2)$) *and* associative, so it runs as a log-depth **parallel scan** rather than a sequential loop. A 228M instance trains **~2× faster** (a larger model than the 176M dense baseline, yet faster), reaches the **same perplexity** (20.76 vs. 20.47), and keeps the brain-swap control directionally ordered `own < swapped < random` with the integrated self-test at 14/14. We stress, however, that *as a single carrier this state is causally small and shrinks with scale* (§4): the dynamics are preserved in direction, not in magnitude. The parallel carrier is the efficient mechanism; the lesson of §4 is to make it the *core mixer* so the magnitude follows.

Self-growth on confirmed saturation.

The faster carrier makes *growth* practical. After pretraining, we continue-train on an instruction-heavy, real-LLM-grade corpus (~8B tokens; 40% instruction/chat from OpenHermes-2.5/SlimOrca/UltraChat/Tulu-3, math from NuminaMath/OpenMathInstruct, world knowledge from Cosmopedia/Wikipedia) to address the weakest axis (instruction-following). The model grows itself, but only on *probe-confirmed saturation*: it spawns a function-preserving copy grown by depth and a copy grown by width, trains each briefly, and commits to growth only if validation loss drops past the validation noise — choosing depth vs. width by whichever helps more. There is no hardcoded size cap; the ceiling is *derived from the data budget* (Chinchilla ≈ 20 tokens/param), and VRAM is the only physical limit. Because the diagonal carrier is d-shaped and unchanged by widening the MLP or adding a block, growth composes with it directly. (Post-training shifts the model off the web-text validation distribution, so checkpoints are saved by recency, not by best validation perplexity — a save-on-best gate would never fire.)

An earlier trigger grew on *any* validation plateau rather than true saturation; an undertrained model accordingly ballooned to `1.4B` parameters far ahead of the tokens needed to fill it (≈28B tokens at 20 tok/param), and its perplexity drifted up — over-growth, not a result. The probe-confirmed criterion above is the fix: capacity is added only when extra capacity demonstrably reduces loss. We therefore treat the 1.4B instance honestly as undertrained-for-its-size, and the self-governing growth rule as the correction.

Serving and memory at scale.

Three engineering properties follow from the diagonal carrier. (i) Inference runs in **bf16**, and generation is `O(T)` at long context: attention stays inside its trained window while the carrier transports cross-window memory, so context length is unbounded without quadratic attention cost. (ii) Full-parameter training of large grown models fits on small GPUs via bf16 + gradient checkpointing + paged 8-bit Adam — no parameter freezing or adapters. (iii) An in-weights *subconscious* fast-weight associative memory writes surprise-gated Hebbian traces, recalls them additively, decays/forgets, and consolidates into the slow weights during a "sleep" phase. This is a learned in-weights memory, **not** retrieval or RAG — there is no external index at recall time.

5. Integration, Self-Calibration, and Autonomy

All faculties and the hybrid language model are wired into a single object, "Pragnosia" (Fig. 2). Three design commitments distinguish it from a conventional model.

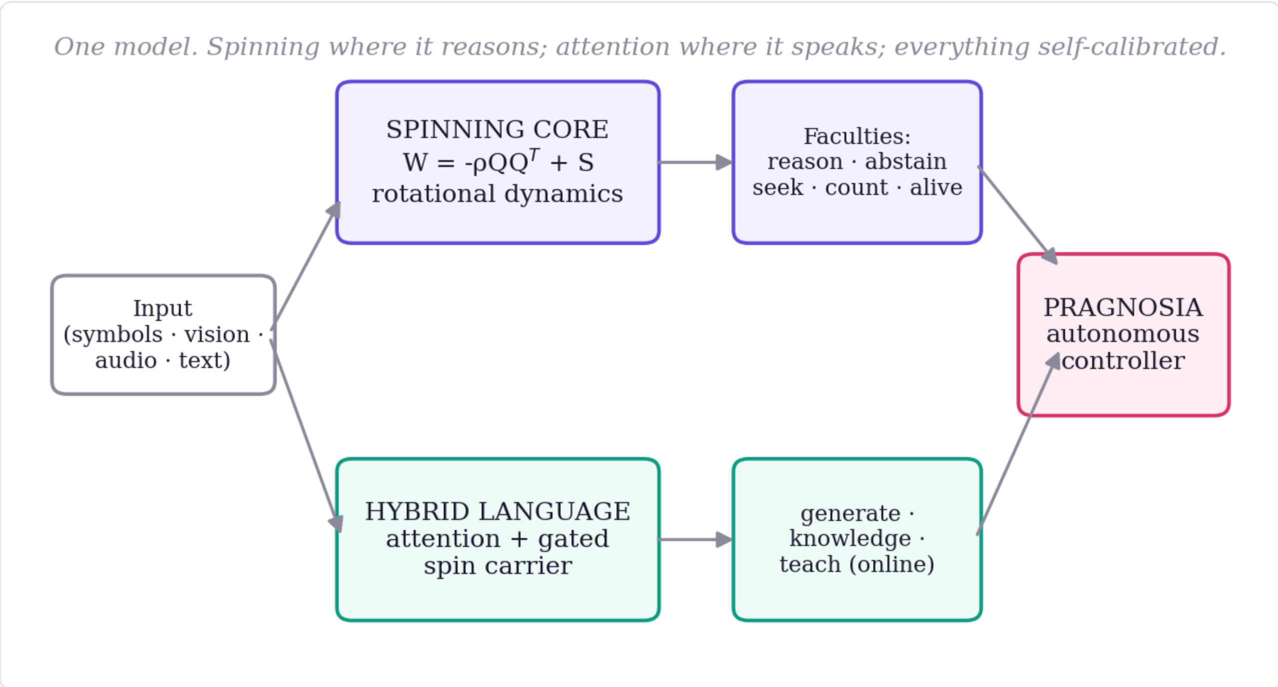

Figure 5. System overview. A shared input feeds the pure-spin reasoning core (top) and the attention+spin-carrier language faculty (bottom); both feed an autonomous controller that decides each action from the model's own signals.

Nothing hardcoded.

Every internal scale is derived from data and re-derived by a `recalibrate()` step as the model grows. Word importance is the self-information `−log p(token)` estimated from the training corpus — function words emerge as low-weight, content words high — with no stopword list. The abstention boundary is the high-percentile of the model's *own* uncertainty on familiar text. The memory-match threshold is read off the corpus's own background similarity distribution. None of these is hand-set.

Surprise-modulated plasticity.

Continual learning is self-paced like neuromodulation. When taught a fact, the brain sets its own learning rate from its own surprise — the prediction error divided by its familiarity boundary — learning hard on surprising input and gently on familiar input, and stopping once the fact ceases to surprise it (Fig. 5). Beyond biological plausibility, gentle steps on familiar input reduce interference. Taught facts persist across sessions; the model recalls them in fresh contexts (e.g., "the capital of Mongolia is" → "Ulaanbaatar") while retaining prior abilities.

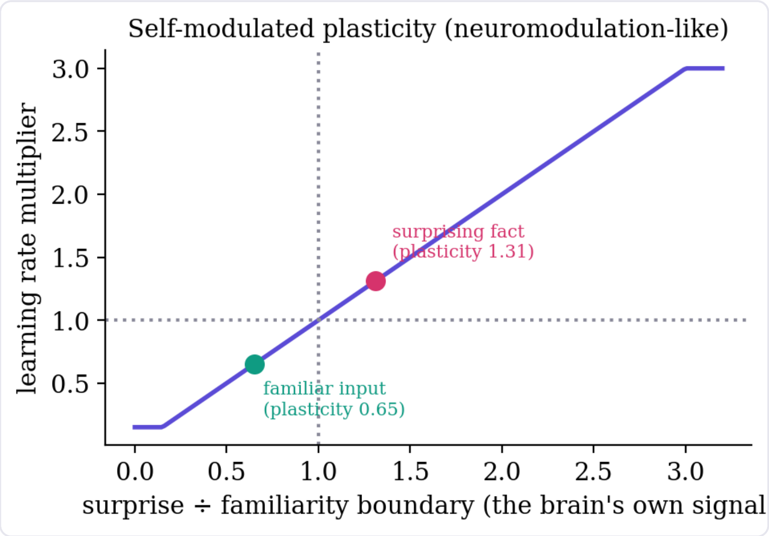

Figure 6. Self-modulated plasticity. The online learning rate is a function of the model's own surprise relative to its self-calibrated familiarity boundary; measured operating points for a surprising and a familiar input are marked.

Autonomous control.

A single controller decides, for any input, whether to answer, seek from memory, abstain, or learn — entirely from its own confidence and curiosity signals, with no keyword rules. Self-knowledge (the model's name and nature) is learned into the weights and recalled by the model's own generation, gated by its own confidence.

6. Experiments and Results

Table 2. The integrated self-test: one model, fourteen checks, all passing. (↓ lower is better.)

Capability	Measured	Criterion	Status
Reasoning ×3 skills (parity / sum / max)	0.92 / 0.90 / 0.93	>0.90	✓
P5 confidence gap (knowable vs. unanswerable)	0.34	>0.30	✓
P6 abstain: known↓ / unknowable	0.19 / 1.00	<0.25 / >0.60	✓
Language parse / generate (held-out)	1.00 / 1.00	>0.80	✓
Seek: query language / answer / random-ctrl↓	1.00 / 1.00 / 0.13	>0.90 / >0.85 / <0.45	✓
Exact accumulation, length 16	1.00	>0.90	✓
Alive loop: final / retention gap↓	1.00 / 0.00	>0.95 / <0.15	✓
Brain-swap (causal): swapped / input / random	0.59 / 0.10 / 0.21	swapped ≫ random	✓

Scaling the language faculty from 0.3M parameters and toy grammars to multi-hundred-million-parameter models on real corpora improves coherence and factual recall, while the full integrated self-test continues to pass — text training provably never touches the reasoning-faculty weights, which we verify after every change.

Trained instances.

The dense-tanh carrier was first verified at 176M parameters (d=1024, 12 layers, vocabulary 16,384), where validation perplexity fell monotonically with no divergence and the full integrated self-test held throughout. The parallel diagonal-complex carrier trains ~2× faster than that 176M baseline and reaches the same perplexity (20.76 vs. 20.47), with the brain-swap control still ordered `own < swapped < random` and the self-test at 14/14 — but in the `single`-carrier configuration, which §4 shows drifts to a causally-negligible side-channel by 1.4B. The self-grown 1.4B instance (val ppl 17.35 after adaptive-lr refinement) is this attention-dominant model; the param-matched *spin-dominant* ablation (§4) is the corrected design, validated at d=512 and awaiting replication at scale. At these scales the models are coherent at the sentence level, write simple correct code (`def add(a,b): → return a + b`), and acquire the *form* of step-by-step reasoning and encyclopedic text; single-step facts and arithmetic are real while multi-hop composition remains weak — the undertraining-for-size signature, expected before the compute-optimal token budget is reached. Continual learning works end-to-end at every scale: the model is taught a new fact, recalls it in a fresh context (e.g. "the capital of Mongolia is" → "Ulaanbaatar"), and retains prior abilities (no forgetting). The one honest weakness is language abstention: confidence-thresholded refusal catches clearly out-of-distribution inputs but is noisy on familiar-phrasing/unknown-content; a trained confidence head (the P6 recipe applied to language) is the intended sharpening.

7. Comparison with Large Language Models

Table 3. Where each paradigm wins. The two are built on opposite bets.

Property	Today's LLMs	Pragnosia
Fluent language / world knowledge	**Strong / vast**	**Decent / limited (small)**
Reasoning depth	**Strong**	**Shallow but real**
Causal-intervention method	**None standard**	**Brain-swap (gates the carrier)**
Knows what it doesn't know	**Partial, bolted on**	**Intrinsic (abstains)**
Learn after training, no retrain	**No (frozen)**	**Yes (persistent)**
Forgetting on update	**Catastrophic**	**Low (replay + plasticity)**
Self-paced learning rate	**Fixed schedule**	**Surprise-modulated**
Hand-set behavior	—	**None (self-derived)**
Size / cost	**Data-center**	**Single laptop GPU**

An LLM is a brilliant but frozen library; Pragnosia is a tiny living mind. The bet is that the properties a frozen model lacks — being alive, honest about ignorance, and causally reasoning — are worth pursuing on a substrate designed for them, even at the cost of present-day breadth.

8. Limitations

We state these plainly, as every claim above carries a number. (i) **The scaled model is built backwards.** Its single spin carrier is causally negligible at 1.4B (0.028% of the loss; §4) because depth/width growth dilutes a fixed-size side-channel. The spin-dominant ablation is the fix, but (ii) **that ablation is small-scale, short-budget, and single-seed** — it shows the intended design wins by ~21% at matched parameters, but it must be replicated at 200M–1B before it is proven; transformers sometimes close such gaps with more scale. (iii) The model is *under-trained for its grown size*: single-step reasoning, arithmetic, and fact recall are real, but multi-hop composition — relational analogy, multi-step deduction, systematic generalization — is weak. The lever is more training tokens, not more parameters or prompting. (iv) Like all small models it can be *confidently wrong* on familiar-but-incorrect facts; its abstention catches the clearly-unknown, not the subtly-wrong. (v) Self-growth must be governed: an earlier plateau trigger over-grew the model to 1.4B; the probe-confirmed-saturation rule fixes this but the resulting large checkpoint still needs the tokens to fill it. (vi) Several non-language faculties remain at toy scale, wired and verified but not yet individually scaled. None of these is hidden; they are the honest frontier.

9. Future Directions and Applications

Roadmap.

The mechanisms are established; the remaining work is scale, data, and maturation of each faculty. *Near-term*: replace the calibrated-threshold abstention with a trained confidence head (the proven two-phase recipe applied to language) for sharper refusal, and — the dominant lever — enlarge the corpus with reasoning-rich data, since reasoning is learned from examples rather than engineered. *Mid-term*: scale the living faculties — continual learning from a handful of facts to thousands with rehearsal-based consolidation, language-mediated seeking against a real retrieval index (RAG-style), and relational vision and audio beyond toy grids and tones. *Long-term*: a single on-device model that runs continuously, learns its user over months without forgetting, abstains rather than fabricates, and can explain its reasoning — an always-on, lifelong, personal mind rather than a frozen oracle.

Applications.

The compelling use cases are precisely those a frozen, cloud-bound model cannot serve. (i) *On-device assistants that learn the user*: a small, private model that continually acquires a person's facts and preferences locally, with no retraining and no data leaving the device. (ii) *Honest assistance in high-stakes domains* (medicine, law, finance): intrinsic abstention — saying "I do not know" and seeking — is safer than fluent fabrication. (iii) *Lifelong-learning robotics and embedded agents*: curiosity-driven online learning from the environment without catastrophic forgetting, within a footprint that fits on the edge. (iv) *Auditable AI*: causally-probeable reasoning (brain-swap) for accountability-sensitive settings. (v) *Adaptive tutoring and personalization*: a model that adapts to an individual learner over time rather than serving a static average. In each case the differentiator is not breadth of knowledge but the trio of properties this substrate is designed for — alive, honest, and causal — at a cost that runs anywhere.

10. Conclusion

Pragnosia is a transformer/SSM hybrid in which a rotational spin recurrence is meant to be the core token-mixer, integrated with faculties for causally-verifiable reasoning, calibrated abstention, seeking, surprise-modulated continual learning, and self-governed growth — all with no hand-set constants. Our most important honest finding is methodological: a single spin carrier *drifts to a causally-negligible side-channel* as the model scales, and the brain-swap intervention is what let us measure that. A param-matched ablation then shows the intended *spin-dominant* design — spin as the core mixer — is the most parameter-efficient (219 vs 279 ppl), so the carrier earns its weights when it is the core. This result is small-scale and single-seed; the next step is to replicate it at 200M–1B and to train the grown checkpoint to its compute-optimal token budget. We release the full system, every measured number, and every limitation.

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